

Python Programming

Chapter 1 Introduction to Python Programming

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Textbook

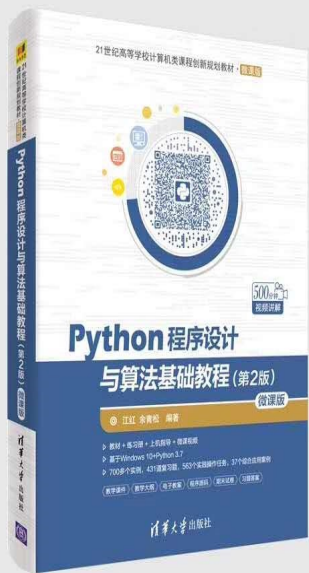
■ Textbook:

■ For All Students: 《Python程序设计与算法基础》。

■ For International Students:

■ Just follow my materials and check here:

- <https://pythonbooks.org/free-books/>



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Arrangement

■ In-Classroom:

- Theory Courses: Week 1 - Week 18
 - Final Presentation: Week ? - Week 18
- Experimental Courses: Week 5 - Week 13
 - About 4 assignments

Composition of Grades

■ Attendance 10 Points

- 10 points for attendance, 7 points for absence once, 4 points for absence twice, and 0 points for absence three times.
- Four absences will result in the cancellation of all grades depending on the situation.

■ In-Class Activities 10 Points

- Pay attention to this.

■ Coursework 40 Points

- Four works, each 10 Points.
- Starting from Week 5.

■ Final presentation 40 Points

- TBD

Section 1

Introduction to Python Programming

1.1 Python Introduction

- How to pronounce this word?

- /'paɪ θ ən/, /'paɪ θ ɑ:n/



1.1 Python Introduction

■ Definition

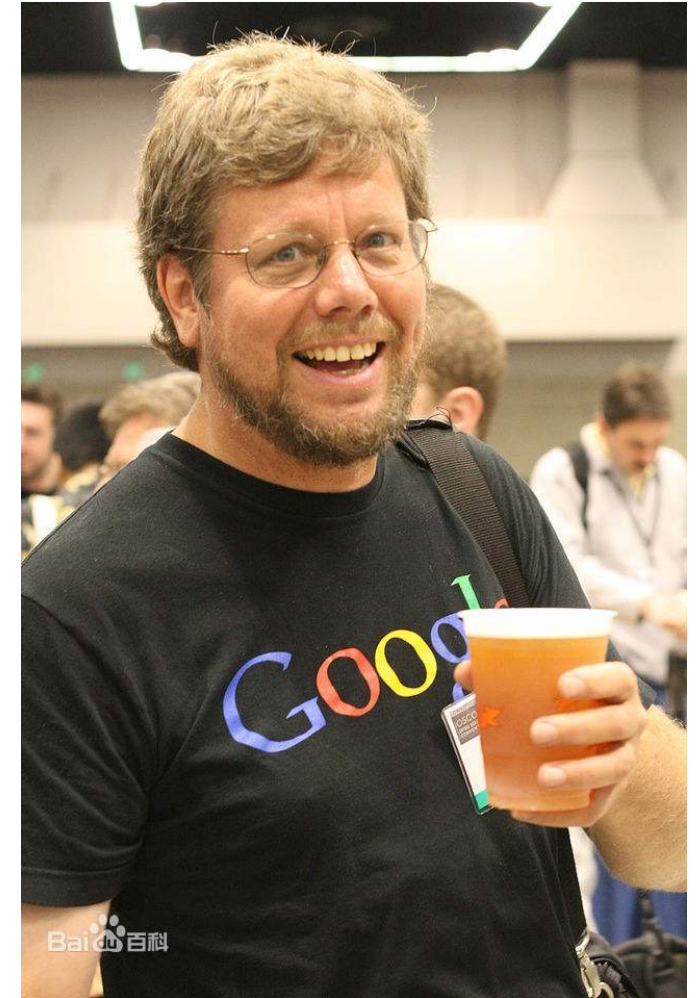
- Python is a **high-level, general-purpose programming** language. Its design philosophy emphasizes code readability with the use of significant indentation.
- Python is an open source language with a large number of libraries.

■ Python Features

- Simple, High-Level, object-oriented, scalable, free and open-source, portable, rich libraries,

■ Python Usages

- Operating system management, scientific computing, web applications, graphical user interface (GUI) development, others (game development, etc.)

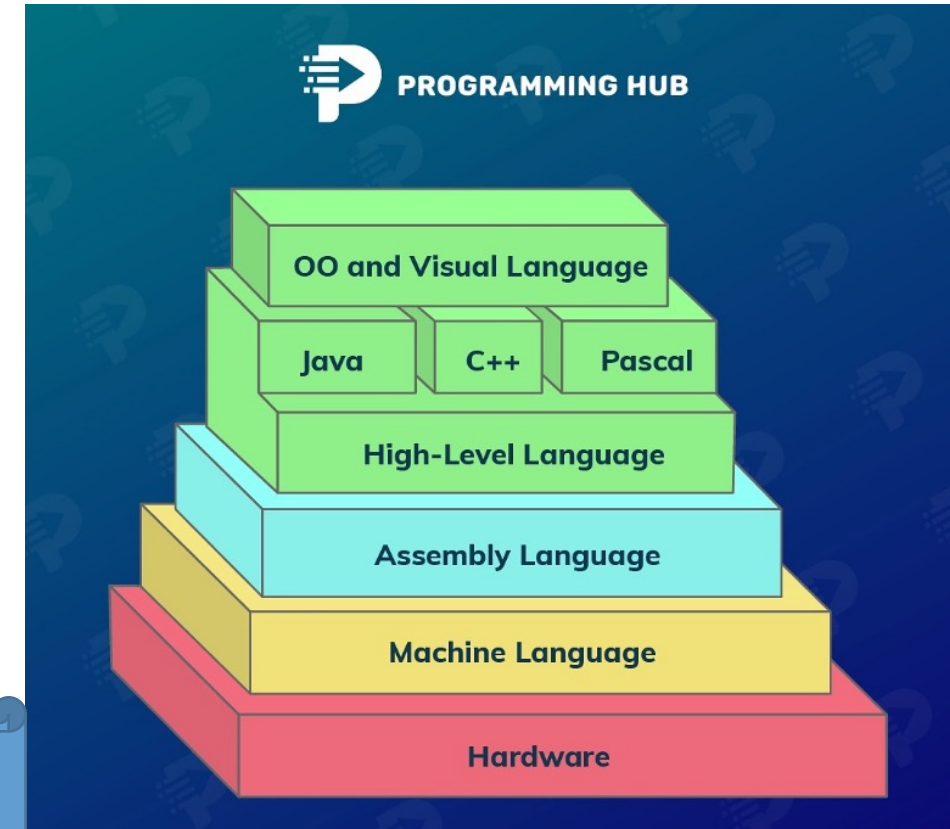


Baidu 百科

1.1 Python Introduction

■ Classification of Programming Languages 1

- Low-level programming language generally refers to machine code or assembly language, which are **closer to hardware and** have a **lower level of abstraction**.
- High-level programming language refers to programming languages with **higher levels of abstraction**, offering better readability and development efficiency. Common examples include C/C++, Java, etc.

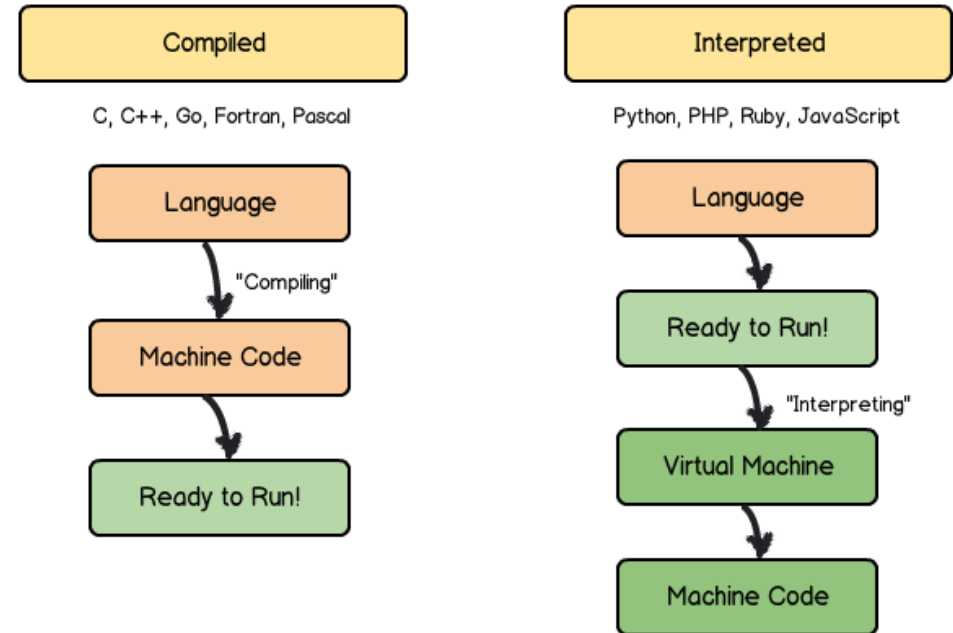


Quiz1: Which is more high-level: C, C++, Python, or Java?
Quiz2: What do we call human languages in computers?

1.1 Python Introduction

■ Classification of Programming Languages 2

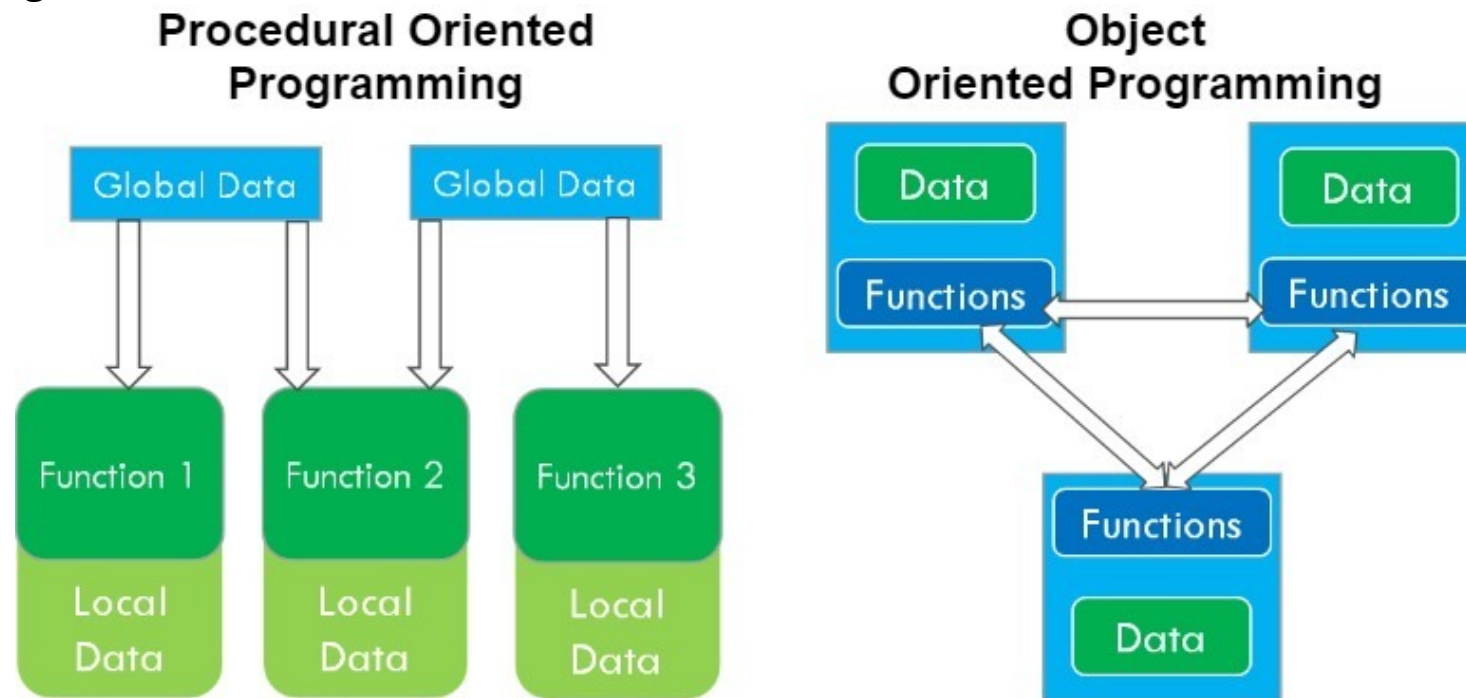
- **Computers cannot directly understand programming languages**, they require conversion into machine languages/codes (X86, ARM, MIPS, RISC-V, etc.) before execution.
- **Compiled language:** Programs must first be compiled into target machine codes.
 - Strong execution efficiency, but poor universality.
- **Interpreted language:** No prior compilation is required; translation to machine language occurs at runtime,
 - Strong universality across platforms, but low efficiency.
 - Scripting languages (a subset of interpreted languages) exist entirely in text form.



1.1 Python Introduction

■ Classification of Programming Languages 3

- **Procedure-Oriented Languages** (e.g., C, Pascal) use functions as their fundamental building blocks. In procedural programming, problems are conceptualized as a series of tasks to be accomplished, with functions executing these tasks.
- **Object-Oriented Languages** (C++, Java, Python) employ objects, classes, and related concepts for program design.



1.1 Python Introduction

■ Key Features of Python

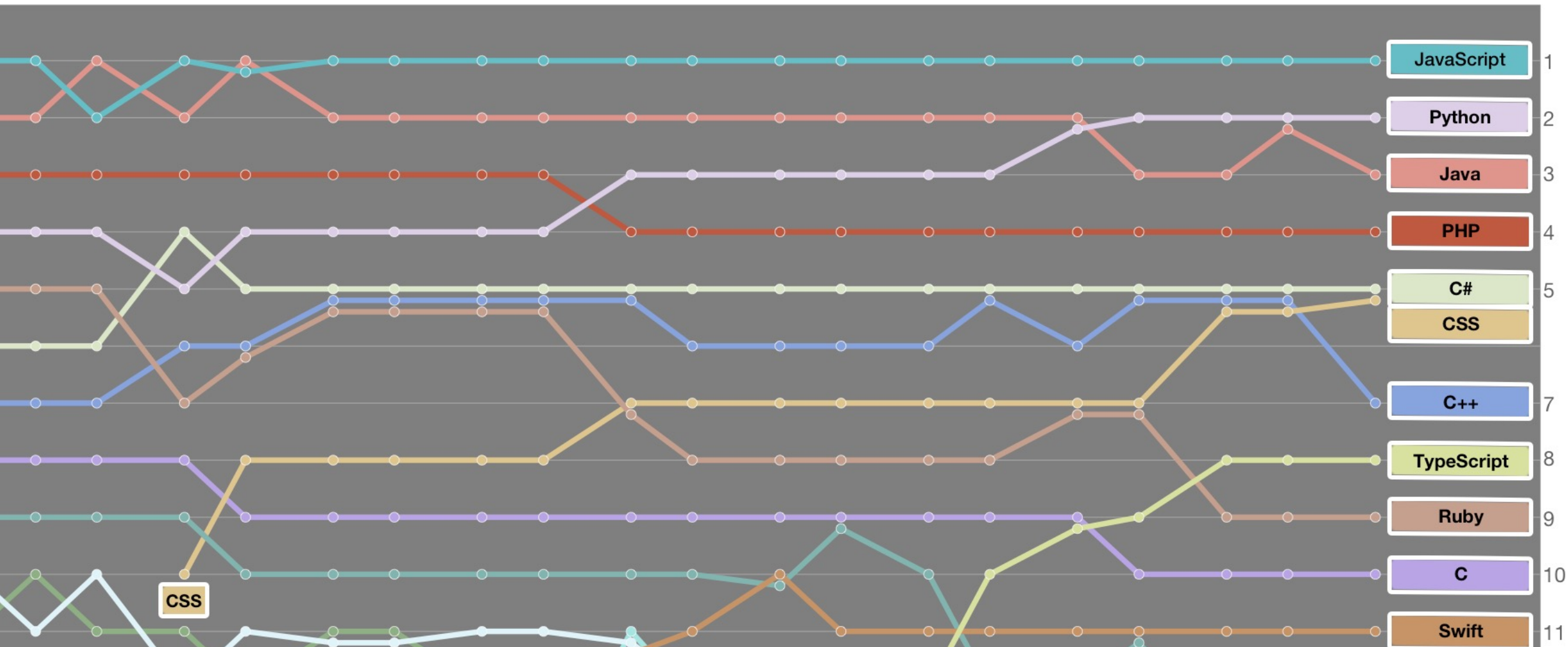
- **Simple:** An interpreted scripting language with straightforward syntax, easy to learn.
- **High-Level:** Python belongs to the category of high-level programming languages.
- **Object-Oriented:** Python is an object-oriented language.
- **Extensibility:** Provides extensive APIs and tools, enabling seamless integration with extension modules written in C/C++.
- **Open-Source:** As open-source software, it allows developers to modify and distribute code freely.
- **Compatibility :** "Write once, run anywhere" across platforms.
- **Rich Libraries:** Boasts a vast ecosystem of extensible libraries.

用Python写出第一行代码后的我
`print "Hello world"`



RedMonk Language Rankings

September 2012 - January 2022



Section 2

Quick Start

1.2 Quick Start

■ Python Versions

- Python has two major versions: Python 2 and Python 3
 - Python 3 was designed without backward compatibility (Primary learning focus)
 - Python 2 provides inadequate Unicode support for non-English languages

■ Python Implementation

- Python interpreter (Translates Python code into machine instructions)

■ Python Integrated Development Environments (IDEs)

- IDLE (Built-in) 、 Spyder、 PyCharm、 Jupyter Notebook、 Eclipse + PydevPlugin、 Visual Studio + Python Tools for Visual Studio、 VSCode

1.2 Quick Start

■ Installing Python

- Download address from the official website : <https://www.python.org/downloads/>
- This course recommends Python 3.7 and above



1.2 Quick Start

■ Using Python

- Open Command Prompt (Windows) or Terminal (macOS/Linux)
- Execute python or python3 to enter interactive mode:

```
-----  
[ (base) orangecat@OrangeCatdeMacBook-Pro .webpack % python  
Python 3.12.7 | packaged by Anaconda, Inc. | (main, Oct 4 2024, 08:22:19) [Clang 14.0.6 ] on darwin  
Type "help", "copyright", "credits" or "license" for more information.  
>>> █
```

1.2 Quick Start

■ Python Environment Variables

- If python or python3 commands fail in Command Prompt (Windows) or Terminal (macOS/Linux), verify your system's environment variables.
- Step 1: Locate Python Installation Directory.
- Step 2: Add to System PATH.

```
[(base) 在你的CMD或者终端直接输入Python] Experiments % Python
Python 3.9.7 (default, Sep 16 2021, 08:50:36)
[Clang 10.0.0 ] :: Anaconda, Inc. on darwin
Type "help", "copyright", "credits" or "license" for more information.
[>>> print("hello world")]
hello world
>>> █
```

1.2 Quick Start

■ Python Interpreter Execution

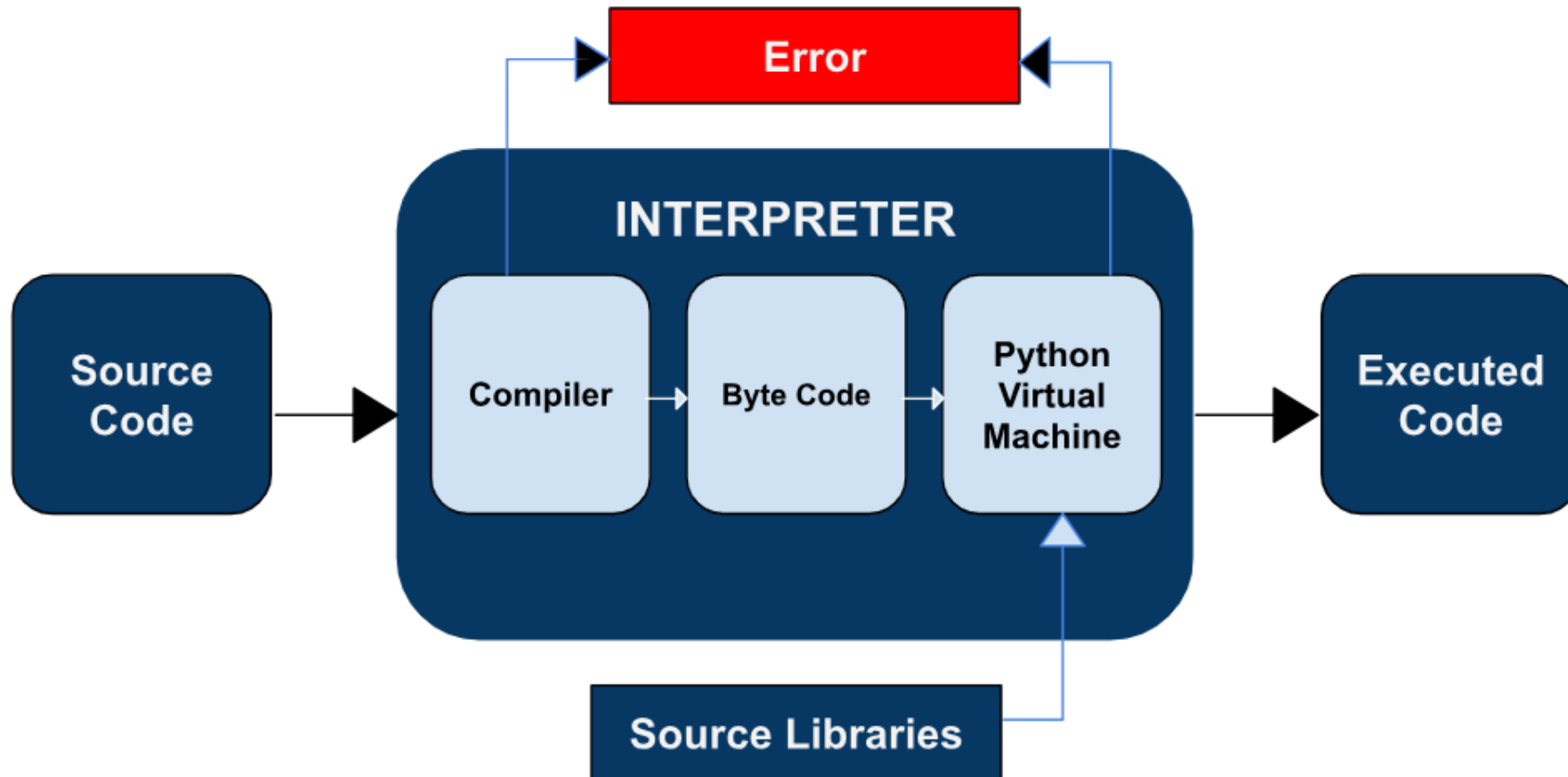
- Code entered directly into the Python interpreter executes dynamically.
- **Commands yielding results will immediately return computation outcomes in the terminal, with all variables preserved during execution.**
- Enter `exit()`, `Control+Z`, and close the terminal directly to end the Python run

```
[>>> 2022+2023
4045
[>>> a = 2023
[>>> a
2023
[>>> exit()
[Ctrl+Z] ^Z
```

1.2 Quick Start

■ Python Code Execution

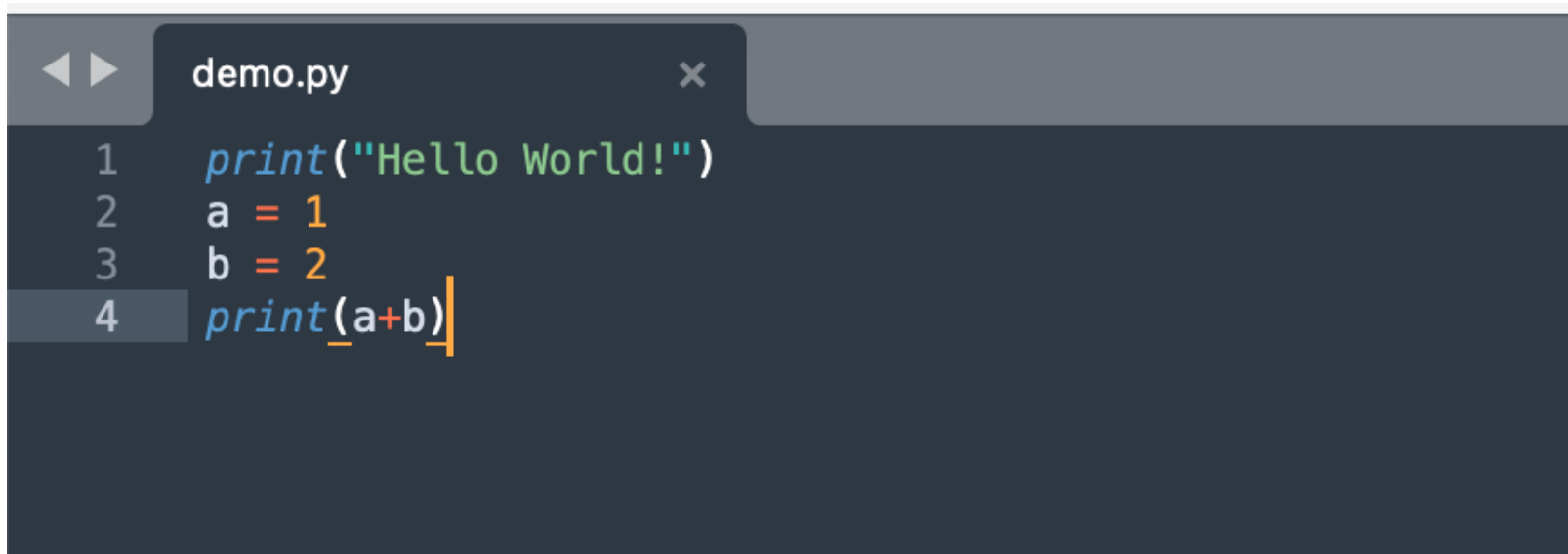
- Python code can be written in a text editor and saved as a “`***.py`” file.
- Run `python “***.py”` in the terminal to execute the script.



1.2 Quick Start

■ Python Source Code Writing and Execution

- Python code can be written in a text editor and saved as a “*.py” file.
- Run python “*.py” in the terminal to execute the script.
- Note: Ensure the file uses UTF-8 encoding.



```
demo.py
1  print("Hello World!")
2  a = 1
3  b = 2
4  print(a+b)
```

```
python demo.py
```

```
Hello World!
3
```

1.2 Quick Start

■ Python Source Code Writing and Execution

- Python code can be written in a text editor and saved as a “*.py” file.
- Run python “*.py” in the terminal to execute the script.
- Note: Ensure the file uses UTF-8 encoding.

```
demo2.py
1  import random
2
3  total_students_num = 50
4
5  number = int(input("请输入今天要点名的学生数量:"))
6
7  candidates = random.sample(range(total_students_num), number)
8
9  for idx, candidate in enumerate(candidates):
10     print("第%d位幸运儿是%d" % (idx+1, candidate+1))
11
12
13
```

```
[sixing@HUAWEIMookX2023 python_demos % python demo2.py
请输入今天要点名的学生数量:5
第1位幸运儿是35
第2位幸运儿是13
第3位幸运儿是5
第4位幸运儿是36
第5位幸运儿是21
[sixing@HUAWEIMookX2023 python_demos % python demo2.py
请输入今天要点名的学生数量:3
第1位幸运儿是30
第2位幸运儿是36
第3位幸运儿是20
[sixing@HUAWEIMookX2023 python_demos % python demo2.py
请输入今天要点名的学生数量:3
第1位幸运儿是40
第2位幸运儿是36
第3位幸运儿是39
sixing@HUAWEIMookX2023 python_demos %
```


1.2 Quick Start

- Python Source Code Writing and Execution - Using Command Line Arguments
 - Import the sys module, then access the i-th argument via “sys.argv[i]” .
 - argv[0] is the name of the executed script, argv[1] is the first parameter, and so forth.

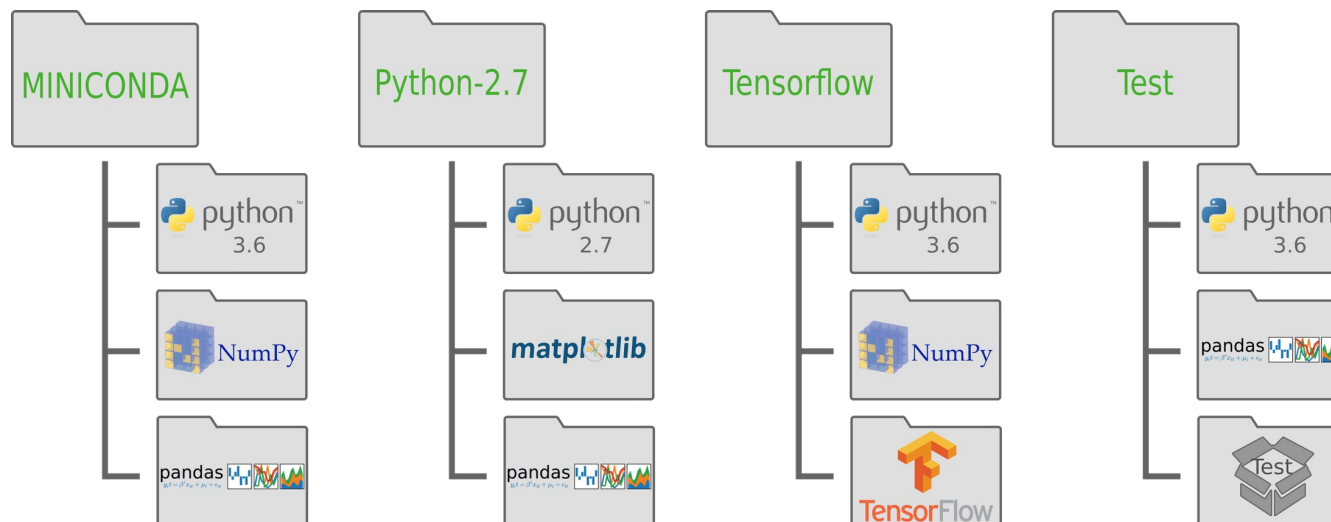
```
demo3.py
1  import random
2  import sys
3
4  print("当前运行文件名为: %s" % sys.argv[0])
5
6  total_students_num = int(sys.argv[1])
7  number = int(sys.argv[2])
8
9  print("今天要点名的学生数量:", total_students_num)
10
11 candidates = random.sample(range(total_students_num), number)
12
13 for idx, candidate in enumerate(candidates):
14     print("第%d位幸运儿是%d" % (idx+1, candidate+1))
15
16
17
```

```
[sixing@HUAWEIMookX2023 python_demos % python demo3.py 50 5
当前运行文件名为: demo3.py
今天要点名的学生数量: 50
第1位幸运儿是 23
第2位幸运儿是 5
第3位幸运儿是 7
第4位幸运儿是 49
第5位幸运儿是 3
[sixing@HUAWEIMookX2023 python_demos % python demo3.py 50 5
当前运行文件名为: demo3.py
今天要点名的学生数量: 50
第1位幸运儿是 25
第2位幸运儿是 11
第3位幸运儿是 37
第4位幸运儿是 42
第5位幸运儿是 4
[sixing@HUAWEIMookX2023 python_demos % python demo3.py 50 3
当前运行文件名为: demo3.py
今天要点名的学生数量: 50
第1位幸运儿是 18
第2位幸运儿是 33
第3位幸运儿是 13
sixing@HUAWEIMookX2023 python_demos %
```

1.2 Quick Start

■ Python Virtual Environments

- Python and its packages exist in multiple versions, with **version-specific dependencies that are prone to incompatibility.**
- For example, Scenario A requires Tensorflow 1.14, while Scenario B necessitates Tensorflow 1.15.
- By creating virtual environments, you **maintain isolated version environments for each application scenario.**



1.2 Quick Start

■ Python Virtual Environments

- venv: Built-in virtual environment tool in Python 3 (not recommended for production).
- **conda (Anaconda): The most widely-used virtual environment manager, adopted in this course.**
- Official Download: <https://www.anaconda.com/>



1.2 Quick Start

■ Using Python Virtual Environments (Terminal Commands)

- **Create Environment:** `conda create --name ENV_NAME python=3.9`
- **Activate Environment:** `conda activate ENV_NAME`
 - Once activated, you are within the virtual environment and can operate Python as you would in a standard environment.
- **Exit Environment:** `deactivate`
- **Delete Environment:** `conda remove - name ENV_NAME --all`
- **List All Environments:** `conda info --envs` or `conda env list`
- **View Installed Packages:** `conda list` (Run within the activated environment)

1.2 Python Language Versions and Development Environments

■ Using Python Virtual Environments (Terminal Commands)

■ Install Packages:

- `conda/pip install xxxxx` (Install xxx package)
- `conda/pip install xxxxx=yyyy` (Install specific version)
- PIP is the default installation package for Python, **and Conda can only be used in Conda (recommended).**

■ Uninstall Packages:

- `conda/pip uninstall xxxx`

■ Learn by Yourself:

- Package Upgrading:.
- **conda/pip Mirror Acceleration, Tsinghua Mirror are generally used, as well as some network acceleration methods .**

1.2 Python Language Versions and Development Environments

■ Python Development Environments

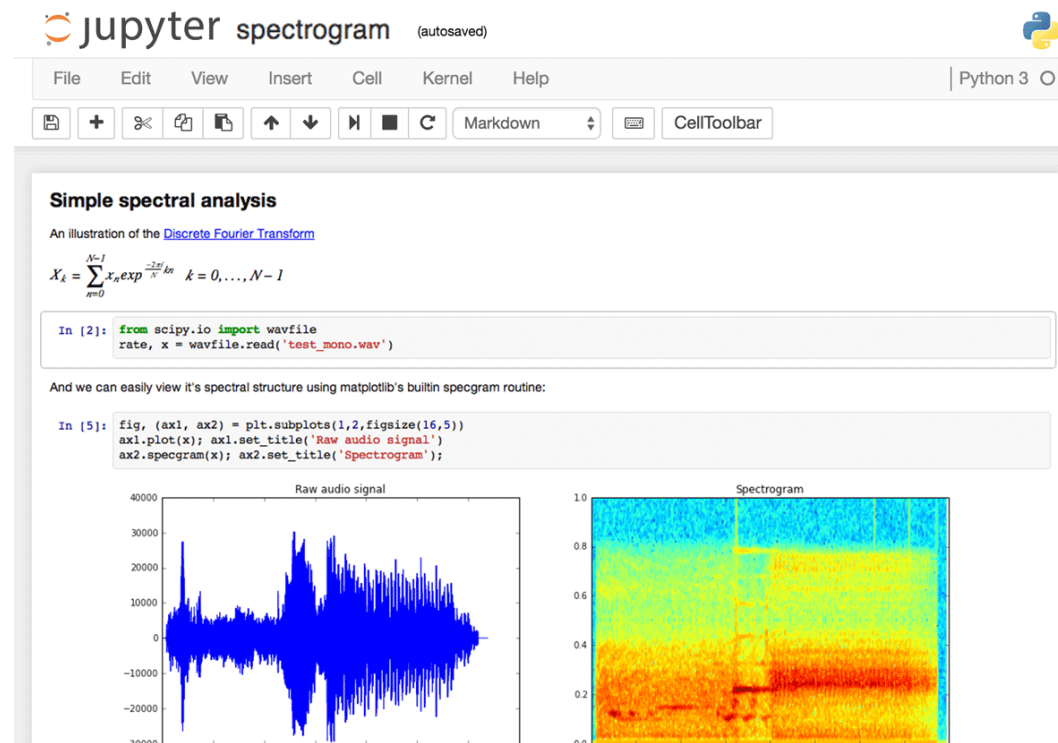
- IDE (Integrated Development Environment): An application that provides comprehensive facilities for software development, typically including: Code editor, Compiler/Interpreter, Debugger, Graphical User Interface (GUI).
- Recommended IDEs for Project Development:
 - PyCharm (Free Community Edition available; Educational licenses for academia)
 - VS Code (Open-source, lightweight & extensible)



1.2 Python Language Versions and Development Environments

■ Python Development Environments

- IDE (Integrated Development Environment): An application that provides comprehensive facilities for software development, typically including: Code editor, Compiler/Interpreter, Debugger, Graphical User Interface (GUI).
- Recommended Interactive IDE for Daily Research & Data Analysis: Jupyter Notebook.



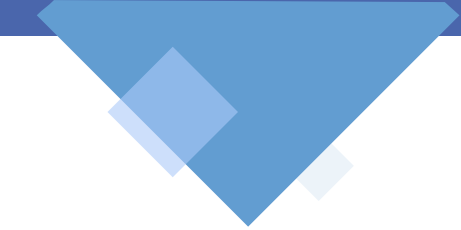
Section 3

Homework Assignment

1.3 Homework

- Homework 1: Try to Install Python !
 - Be sure to have Hello World before the next class!
- Try it: Use DeepSeek !





Thanks !